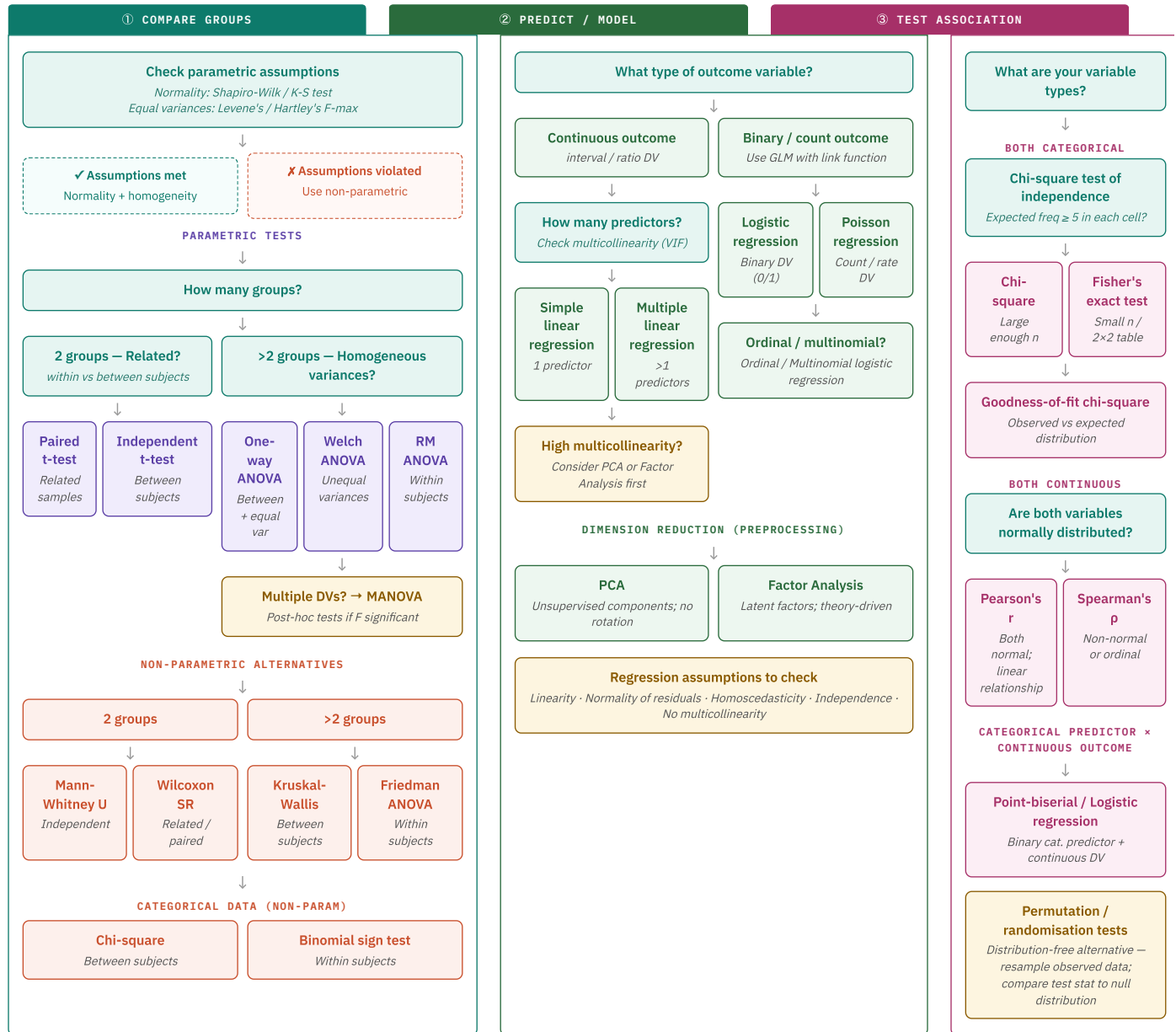


Statistical Test Selection

Based on: BRSM Lectures, Hypothesis Testing, ANOVA, Non-parametric Tests, GLM & Regression

START – What is your research goal?



Running many tests? → Multiple Comparisons Correction

Bonferroni: α / k (conservative) | FDR / Benjamini-Hochberg (less conservative) | Holm-Bonferroni (step-down)

Bayesian alternative (any test)

Prior belief + likelihood → posterior | Bayes Factor (BF): evidence for H_1 vs H_0 | Credible intervals instead of CIs

All tests share the same logic

Define H_0 and H_1 → Set α (typically 0.05) → Calculate test statistic → Obtain p-value → Report effect size (Cohen's d, η^2 , r) → Check statistical power

LEGEND

Decision / question Parametric test Non-parametric test Regression / GLM Categorical / association Caution / correction needed Bayesian framework